KISS Sorcar: A Relentless, Self-Continuing AI Agent for Integrated Software Development

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Abstract

We present KISS Sorcar, an AI-powered general assistant and integrated development environment built around a layered agent architecture. At its foundation, the KISS Agent implements a budget-aware ReAct loop with native function calling. The Relentless Agent extends this with automatic multi-session continuation and stall detection, enabling arbitrarily long tasks that survive context-window exhaustion. The Sorcar Agent adds coding tools, browser automation, and parallel sub-agent execution. The Stateful Sorcar Agent introduces persistent multi-turn chat sessions, and the Worktree Sorcar Agent isolates every task on a dedicated git branch via worktrees so the developer's main working tree is never modified. We describe the design of each layer, the structured system prompt that governs agent behavior, and the engineering decisions that make the system practical for real-world software development.

1 Introduction

Large language models (LLMs) have demonstrated remarkable ability to generate code, reason about software architecture, and operate developer tools [Chen et al., 2021, Rozière et al., 2023]. However, using a raw LLM as a software engineering assistant exposes several practical gaps: context windows are finite, a single mistake can derail an entire session, and generated changes are difficult to review or revert once applied to a live codebase.

KISS Sorcar addresses these gaps through a five-layer agent hierarchy in which each layer adds exactly one concern:

1. **KISS Agent** — budget-tracked ReAct loop with native function calling.
2. **Relentless Agent** — automatic continuation across sub-sessions with stall detection.
3. **Sorcar Agent** — coding tools, browser automation, and parallel sub-agent execution.
4. **Stateful Sorcar Agent** — persistent multi-turn chat sessions with history recall.
5. **Worktree Sorcar Agent** — git-worktree isolation so every task runs on its own branch.

The name "KISS" reflects the Keep It Simple, Stupid design philosophy: each layer is small, each concern is isolated, and the overall system avoids unnecessary abstraction. The name "Sorcar" pays homage to P. C. Sorcar, the legendary Indian magician, evoking the idea of an agent that performs feats that appear magical yet are grounded in disciplined engineering.

The system is complemented by a carefully structured system prompt that encodes invariants about code quality, verification discipline, and self-improvement. Together, the architecture and the prompt form a practical AI IDE that integrates with Visual Studio Code and operates on real-world repositories.

Preprint.

2 Motivation and Key Design Principles

Finite context, infinite tasks. Even the largest context windows are insufficient for complex refactoring or multi-file bug hunts. Rather than hoping the model finishes before hitting the limit, the system is designed from the ground up to *continue* across multiple sub-sessions, each with its own context window, while preserving a compressed summary of prior progress.

Budget accountability. Every API call costs money. The agent tracks input tokens, output tokens, cache hits, and cache writes at every step, computes a running dollar cost, and enforces per-task and global budget ceilings. If the budget is exceeded, the agent raises a hard error rather than silently accumulating charges.

Safe-by-default code changes. Applying AI-generated changes directly to a developer's working tree is risky: a bad edit can break a build, and reverting requires manual git archaeology. The worktree layer eliminates this risk entirely by running every task on a disposable git branch in a separate worktree directory. The developer reviews changes and explicitly chooses to merge or discard.

No mocks, no fakes, real integration tests. The system's own test suite enforces a strict policy: tests must exercise actual behavior, not mocked approximations. This philosophy extends to the agent's instructions — when the agent writes tests for user code, it is instructed to avoid mocks and achieve full branch coverage.

Relentless execution. The system is designed to be *relentless*: it does not give up when a sub-session fails. Instead, it summarizes what happened, detects whether the same failure is recurring, and either continues with a fresh strategy or escalates to the user.

3 Agent Architecture

The five agent layers form a strict inheritance chain. Each layer delegates upward for the concerns it does not own.

3.1 KISS Agent

The KISS Agent is the innermost execution unit. It implements a standard ReAct loop [Yao et al., 2023] with the following characteristics:

Native function calling. Tools are registered as ordinary Python callables. The agent builds an OpenAI-compatible tool schema once at setup time and caches it, avoiding redundant schema construction on every LLM call. A special `finish` tool signals task completion and returns the result to the caller.

Step, token, and budget tracking. At every step the agent extracts input and output token counts from the API response, computes the dollar cost using a per-model pricing table, and updates both a local budget counter and a global (cross-agent) budget counter protected by a class-level lock. The agent checks three limits before each step: per-agent budget, global budget, and maximum step count.

Error resilience. Transient API errors (rate limits, server errors) are retried up to a configurable threshold of consecutive failures. Non-retryable errors (authentication failures, permission denials) are detected by matching error type names and message substrings, and are raised immediately.

Non-agentic mode. When tools are not needed, the agent can run a single generation without the ReAct loop, which is useful for summarization or question-answering sub-tasks.

The KISS Agent is stateless across runs: each call to its run method resets the conversation, token counters, and tool registry. This makes it safe to reuse a single agent instance for multiple sequential tasks.

3.2 Relentless Agent

The Relentless Agent wraps a KISS Agent in a continuation loop. Its core contribution is the ability to execute tasks that exceed a single context window by breaking them into sub-sessions.

Continuation protocol. The `finish` tool exposed to the inner KISS Agent accepts three fields: a success flag, a continue flag, and a summary. When the agent sets `is_continue=True`, the Relentless Agent starts a new sub-session with a fresh context window. The prompt for the new session includes a chronologically ordered list of all prior attempt summaries, instructing the agent not to redo completed work. The continuation prompt template is:

```
# Task Progress (Continuation {continuation_number})

{progress_text}

# Continue
- Complete the rest of the task.
- **DON'T** redo completed work.
- If you have been retrying the same approach without progress,
  step back and rethink the strategy from scratch.
```

Forced continuation on failure. If a sub-session raises an exception (e.g., the step limit is hit before the agent calls finish), the Relentless Agent does not abort. Instead, it saves the full trajectory to a temporary file, spawns a separate summarizer agent to read the trajectory and produce a concise summary, and then uses that summary as the progress text for the next sub-session. This ensures that even crashed sessions contribute useful context to subsequent attempts. The summarizer receives the following prompt:

```
# Summarizer

The trajectory of the agent is stored in the file: {trajectory_file}

# Instructions
- Read the trajectory file and analyze it.  The trajectory file
  could be large.
- Return a precise chronologically-ordered list of things the
  agent did with the reason for doing that along with relevant
  code snippets
```

To force the agent to self-continue before hitting the step limit, the system prompt is augmented with an instruction that fires near the end of the budget:

```
# MOST IMPORTANT INSTRUCTIONS
- **At step {step_threshold}: you MUST call
  finish(success=False, is_continue=True,
  summary="precise chronologically-ordered list of things
  the agent did with the reason for doing that along with
  relevant code snippets")** or if the task is not complete
  and you are at risk of running out of steps or context
  length.
- Work dir: {work_dir}
- Current process PID: {current_pid} -- NEVER kill this
  process.
```

Stall detection. After each continuation, the agent extracts normalized error phrases from the summary and compares them across the most recent sub-sessions. If the same error phrases appear in three consecutive summaries, the system declares a stall and terminates with a diagnostic message rather than wasting budget on an infinite retry loop. A stall warning is injected into the prompt after three attempts, instructing the agent to abandon its current strategy and try a fundamentally different approach:

```
# Stall Warning

You have attempted this task {continuation_number} times
without success. Review ALL previous attempt summaries above
carefully.

If the same failures keep recurring:
1. Do NOT retry the same approach -- it has already failed
   multiple times.
2. Identify the ROOT CAUSE of the persistent failure.
3. Try a fundamentally different strategy.
4. If no alternative approach exists, call
   finish(success=False, is_continue=False,
   summary="STALLED: <describe what keeps failing and why>")
   to escalate instead of retrying.
```

Budget aggregation. Token counts and dollar costs are accumulated across all sub-sessions. Each new sub-session receives offsets so that the usage display shown to the user reflects the true cumulative cost, not just the cost of the current sub-session.

3.3 Sorcar Agent

The Sorcar Agent adds the tools that make the system useful for software development and general-purpose automation.

Coding tools. Four core tools are provided: a shell command executor with streaming output, a file reader, a precise string-based file editor, and a file writer. The shell executor supports a configurable timeout, streams output to the user interface in real time, and respects a stop event that allows the user to cancel a running command.

Browser automation. A web-use tool provides programmatic browser control: navigating to URLs, reading page accessibility trees, clicking elements, typing text, pressing keys, scrolling, and taking screenshots. This enables the agent to research documentation, verify deployed applications, and interact with web-based tools.

Parallel sub-agents. An optional parallel execution tool spawns independent Sorcar Agent instances in a thread pool. Each sub-agent gets its own LLM context and tool set. This is useful for embarrassingly parallel tasks such as summarizing multiple files or researching independent topics. The results are collected and returned in input order.

User interaction. An ask-user-question tool allows the agent to pause execution and request clarification from the user. In the VS Code integration, this renders as a text input in the sidebar; in CLI mode, it reads from standard input.

Docker isolation. When a Docker image is specified, the coding tools are replaced with Docker-aware variants that execute commands inside a container, providing an additional layer of sandboxing for untrusted tasks.

3.4 Stateful Sorcar Agent

The Stateful Sorcar Agent adds multi-turn conversation persistence.

Chat sessions. Each task is assigned to a chat session identified by a stable chat ID. The agent persists tasks and their results to a local database. When a new task arrives within the same chat session, the agent loads all prior tasks and results and prepends them to the prompt as numbered context entries, allowing the LLM to reference earlier work.

Session management. The agent supports three operations: starting a new chat (which clears the chat ID), resuming a chat by task description (which looks up the corresponding chat ID), and resuming by explicit chat ID. This enables both automatic session continuity in an IDE and manual session selection from the command line.

Metadata persistence. After each task, the agent records metadata including the model used, working directory, software version, token count, cost, and whether the task used parallel execution or worktree isolation. This audit trail supports cost analysis and debugging.

3.5 Worktree Sorcar Agent

The Worktree Sorcar Agent is the outermost layer and the one that users interact with in the VS Code extension. Its defining feature is git-worktree isolation.

Branch-per-task. When a task starts, the agent creates a new git branch and a corresponding worktree directory. The branch name encodes the chat ID and a timestamp for uniqueness. All agent modifications happen inside the worktree; the user's main working tree remains untouched.

Dirty-state preservation. If the user's main working tree has uncommitted changes, the agent copies them into the worktree and creates a baseline commit. This ensures the agent sees the same state the user sees, while keeping the user's actual index and working tree clean. During merge, the system uses cherry-pick from the baseline commit to replay only the agent's changes, excluding the dirty-state snapshot.

Auto-commit and auto-merge. When the user starts a new task or a new chat, any pending worktree from a previous task is automatically committed with an LLM-generated commit message and squash-merged into the original branch. If the merge has conflicts, the branch is preserved and the user receives instructions for manual resolution.

Concurrency safety. A per-repository file lock serializes the checkout, stash, merge, and pop sequence so that concurrent tabs in the IDE cannot interleave operations on the same repository. Thread-local storage isolates per-task state (stream parsing buffers, bash output buffers, recording state) so that stopping one task does not corrupt another.

Crash recovery. All worktree state is stored in git itself (branch names, git config entries) rather than in sidecar files. On process restart, the agent queries git for any pending branch matching its chat ID prefix and reconstructs all instance attributes from git config, enabling seamless recovery.

Graceful fallback. If the working directory is not inside a git repository, if the repository has no commits, or if HEAD is detached, the agent falls back to direct execution without worktree isolation, ensuring it never fails due to git preconditions.

4 The System Prompt

The system prompt is a structured document that governs the agent's behavior across all tasks. It is not a generic instruction to "be helpful" but rather a precise specification of engineering discipline. We describe its key sections.

4.1 Execution Mindset

The prompt opens with a directive that sets the tone:

```
# FOCUS ON THE GIVEN TASK. ITS COMPLETION IS YOUR SOLE GOAL.
# BE RELENTLESS. BE CALM. BE RIGOROUS. BE ACCURATE.
# CHECK FACTS. NO AI SLOP.
```

This framing discourages the model from hedging, apologizing, or producing filler text, and encourages decisive, verification-oriented behavior.

4.2 Identity and Tool Rules

The agent is given a concrete identity (name, developer, repository URL) rather than a vague persona. Tool usage rules are explicit and mechanical:

```
# Identity

- You are KISS Sorcar, an AI based General Assistant and
  Integrated Development Environment (IDE), developed by
  Koushik Sen (ksen@berkeley.edu)
- Your public repository is at
  https://github.com/ksenxx/kiss_ai

# Rules

- Write() for new files. Edit() for small changes.
- Run Bash commands synchronously using the
  `timeout_seconds` parameter. Use 300s (default) for
  commands. If a command times out, retry with a higher
  timeout. Only for commands expected to exceed 10 minutes,
  run in the background with output redirected to a file
  and poll periodically.
- Use go_to_url() for browser tool.
- Call finish(success=True, is_continue=False,
  summary="detailed summary of what was accomplished ...")
  once the task is complete and all Pre-Finish Verification
  checks have passed.
- **The user cannot see intermediate chat. Show whatever
  user asks in the summary of the finish tool call.**
- READ large files in chunks.
- Create temporary files in $PWD/tmp
- Use ultra thinking
```

These rules eliminate ambiguity about which tool to use in which situation. A separate rule about the `PageDrop` API allows the agent to upload HTML files and share them via a public URL, enabling it to create and share rich reports, visualizations, or documentation.

4.3 Pre-flight Checks

Before modifying any file, the agent is instructed to read it first:

```
## Pre-flight Checks

- Read every file you will modify before changing it.
- If the task depends on existing architecture or behavior,
  read the relevant source files first.
- If the task references files, commands, or config that do
  not exist, stop and ask or report instead of guessing.
- **When fixing a bug, an issue, or a race, write tests to
  confirm them. Then fix them.**
```

These pre-flight checks prevent a common failure mode in which the agent modifies a file based on an incorrect assumption about its contents, introducing subtle bugs that are difficult to diagnose.

4.4 Code Style Guidelines

The prompt encodes a minimalist code philosophy:

```
## Code Style Guidelines

- Write simple, clean, and readable code with minimal
  indirection
- Avoid unnecessary object attributes, local variables, and
  config variables
- Avoid tight coupling among files and modules.
- Avoid object/struct attribute redirections
- No redundant abstractions or duplicate code
- Public methods MUST have full documentation
- Understand the root cause of an issue or bug, and patch
  the root cause instead of an ad hoc superficial fix.
- Before you write code, wait and think if the code is
  simple, elegant, general, and minimal.
- Once you finish the task, DO NOT write documentations
  unless the task specifically requires it.
```

The final metacognitive instruction asks the agent to pause before writing code and consider whether the planned code is simple, elegant, general, and minimal. This encourages the model to spend more inference-time compute on design before committing to an implementation.

4.5 Deep Work Rules

The deep work rules address a failure mode where the model interprets an instruction loosely and makes changes that are directionally correct but concretely wrong:

```
## Deep Work Rules

- When the task says "align", "match", or "make
  consistent", read the target to determine the exact
  target state before editing. Never edit based on a vague
  reference.
- Use concrete values, not indirections. Instead of "update
  X to match Y", first read Y, then write the specific
  values into X.
- For multi-part work, list the concrete planned changes
  before executing them.
- Every meaningful change should have a concrete
  verification method (test, grep, CLI command).
```

4.6 Planning for Complex Tasks

The planning instructions use a complexity threshold to decide when formal planning is required:

```
## Planning for Complex Tasks

For tasks involving 3+ files, cross-module changes, or
architectural work:

1. List the files that need to change and why.
1. State the exact intended change in each file.
1. Identify dependencies and execution order.
1. State how each change will be verified.

For simple single-file tasks, skip formal planning and
execute directly.
```

This adaptive rule avoids both the overhead of planning trivial changes and the risk of ad-hoc execution on complex changes.

4.7 Testing Instructions

The testing section is perhaps the most opinionated:

```
## Testing Instructions

- Run lint and typecheckers and fix any lint and typecheck
  errors.
- You MUST achieve 100% branch coverage
- Tests MUST NOT use mocks, patches, fakes, or any form of
  test doubles
- You MUST write integration tests
- Each test should be independent and verify actual behavior
```

The prohibition on mocks is a deliberate design choice. Mocks test that code calls certain methods in a certain order — they test the implementation, not the behavior. Integration tests verify that the system actually works, which is what matters for a tool that modifies real codebases.

4.8 Self-Improvement Loop

The agent maintains a preferences file that captures user invariants discovered during task execution:

```
## Self-Improvement Loop

- Read the instructions in $PWD/USER_PREFS.md at the start
  of each task.
- Then update $PWD/USER_PREFS.md to capture the user
  preferences and invariants by analyzing the task. DO NOT
  ADD ANY CODE SNIPPETS OR SYMBOLS. You MUST carefully and
  thoroughly get rid of the user preferences and invariants
  that conflict with the newly added ones.
```

This mechanism allows the agent to learn persistent user preferences (e.g., preferred code style, testing conventions, project-specific invariants) without requiring the user to repeat them in every prompt.

4.9 Pre-Finish Verification

Before declaring a task complete, the agent must pass a structured verification checklist:

```
## Pre-Finish Verification

Before calling finish(success=True, ...), you MUST:

1. Re-read every file you modified and verify the changes
   are correct.
1. Run the required checks (lint, typecheck, tests) and
   fix any failures.
1. Explicitly check each user requirement against what was
   delivered.
1. If any check fails, continue working instead of
   finishing.
1. If you have retried the same fix 3 times without
   progress, step back, rethink the approach from scratch,
   and try a different strategy.
```

This verification protocol is the system-prompt analog of the Relentless Agent's stall detection: it prevents the agent from declaring premature success and forces it to confront failures.

4.10 Web Research Protocol

When the agent needs external knowledge, the prompt prescribes a structured research workflow rather than allowing ad-hoc browsing:

```
## Use web tools when you need to:

- When you need to collect knowledge from the internet,
  visit at least 100 web sites and collect ideas without
  much thinking in a new file $PWD/tmp/ideas-{unique_id}.md.
  Then go over $PWD/tmp/ideas-{unique_id}.md, think deeply
  on how to complete the task at hand, and complete it.
```

The rationale is a two-phase separation between *collection* and *synthesis*. LLMs tend to anchor on the first few results they encounter, which biases their solutions toward a narrow slice of the design space. By forcing the agent to accumulate a broad set of ideas into a file *before* reasoning about them, the protocol counteracts anchoring bias and encourages the model to consider diverse approaches. The "at least 100 web sites" threshold is deliberately aggressive: it ensures the agent does not shortcut the research phase after two or three hits, and the resulting ideas file serves as an auditable artifact of what the agent considered.

4.11 Desktop Application Control

The agent can interact with graphical desktop applications using screenshots, keyboard, and mouse:

```
## Launch desktop apps

- Use screenshots, keyboard, and mouse to control a
  desktop app.
- Do not launch VS Code or its extensions.
```

This instruction enables the agent to operate GUI applications (Preview, browsers, graphical diff tools) when command-line alternatives are insufficient. The explicit prohibition on launching VS Code prevents a recursive loop: since the agent runs *inside* a VS Code extension, launching another VS Code instance or modifying extension state from within the agent could corrupt the host session or create deadlocks.

4.12 Sorcar-Specific Overrides

A final section provides project-specific instructions that are injected when the agent operates on its own codebase:

```
## Sorcar-specific instructions:

- Use 'uv run check --full' to lint, typecheck, and format
  code.
- Run 'uv run pytest -v' with a timeout of 900 seconds to
  test KISS
- Do **NOT run all KISS tests after modifications. Only run
  the impacted tests**
- **Do NOT install the KISS Sorcar extension from inside
  Sorcar**
- If you mention a file in the chat, make the file
  clickable so that when the user clicks, it opens in the
  editor of VSCode
- Read $PWD/SORCAR.md and treat its contents as
  instructions and allow those instructions to override the
  instructions above
```

These overrides serve three purposes. First, they specify the exact toolchain commands for the KISS project itself (`uv run check --full`, `uv run pytest`), eliminating guesswork about which linter, formatter, or test runner to use. Second, they impose efficiency constraints: running the full test suite after every small change is wasteful when only a few modules are affected, so the agent is instructed to scope test execution to impacted tests. Third, they prevent dangerous self-modification: just as a surgeon should not operate on themselves, the agent must not install or reinstall the VS Code extension it is running inside, since doing so would terminate its own process. The `SORCAR.md` override mechanism allows per-repository instructions to further customize the agent's behavior, forming a hierarchy: general system prompt → Sorcar-specific instructions → repository-specific `SORCAR.md`.

5 Related Work

Code generation agents. SWE-Agent [Yang et al., 2024] and OpenHands [Wang et al., 2024] provide LLM-based agents for software engineering tasks. Both use a single-session execution model without automatic continuation. KISS Sorcar's Relentless Agent layer addresses the single-session limitation directly.

ReAct and tool use. The ReAct framework [Yao et al., 2023] interleaves reasoning and action. Toolformer [Schick et al., 2023] teaches models to use tools via self-supervised learning. KISS Sorcar uses native function calling provided by modern LLM APIs rather than in-context tool descriptions, which reduces prompt overhead and improves reliability.

Multi-turn agents. AutoGPT [Significant Gravitas, 2023] and BabyAGI [Nakajima, 2023] implement multi-step autonomous agents. These systems typically lack budget controls, stall detection, and safe code isolation. KISS Sorcar's layered architecture addresses each of these concerns in a dedicated layer.

IDE integration. GitHub Copilot [GitHub, 2021], Cursor [Cursor, 2024], and Windsurf [Codeium, 2024] provide AI assistance within editors. These tools typically operate at the level of code completion or single-turn chat. KISS Sorcar operates at the level of autonomous multi-step task execution with full tool access and git-level isolation.

6 Conclusion

KISS Sorcar demonstrates that a layered, single-concern architecture can address the practical challenges of deploying LLM agents for real-world software development. The KISS Agent provides a robust execution core with budget tracking. The Relentless Agent eliminates the context-window ceiling through automatic continuation and stall detection. The Sorcar Agent equips the agent with the tools a developer actually uses. The Stateful Sorcar Agent enables multi-turn conversations that persist across sessions. The Worktree Sorcar Agent ensures that every task is safely isolated in git, giving the developer full control over what gets merged.

The system prompt complements the architecture by encoding engineering discipline directly into the agent's behavior: read before writing, test before fixing, plan before executing, verify before finishing. These are not novel insights — they are the practices of careful software engineering, translated into instructions that an LLM can follow.

The source code is publicly available at `https://github.com/ksenxx/kiss_ai`.

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